UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 6B – Design of Displays and Controls II

EIN3241: Human Factors and Ergonomics
Spring 2026

March 27, 2026

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Admin Issues

- Project:
  - Progress Report due on Monday, March 30
- Quiz:
  - Wednesday, April 1
  - Lectures 5F, 5G, 6A, and 6B

What did we learn last time?

Control Principles

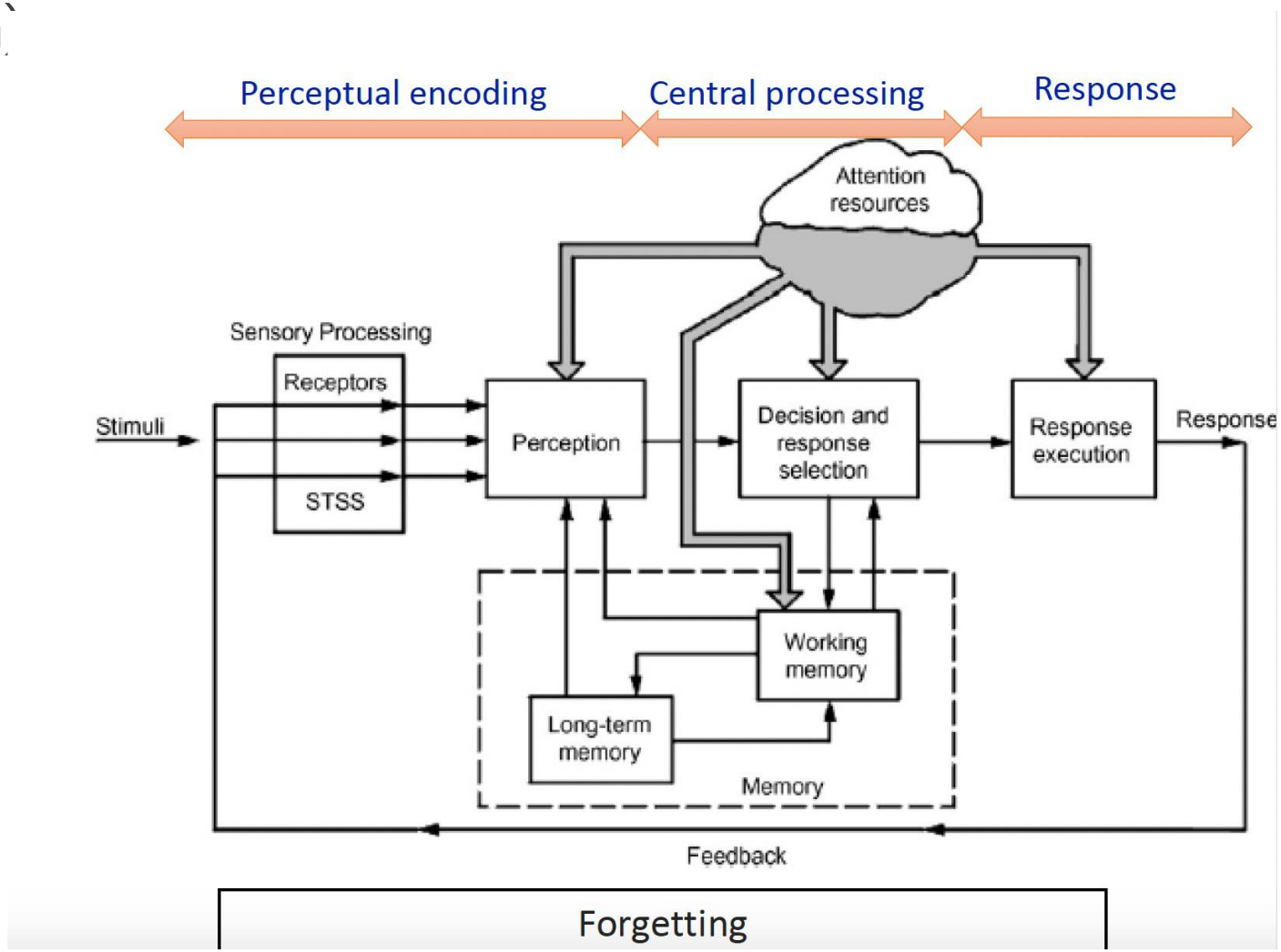
Attention	Perceptual	Memory	Mental Model	Response Selection
1. Proximity compatibility	3. Make accessible	7. Knowledge in the world	9. Location compatibility	11. Avoid accidental activation
2. Avoid resource competition	4. Make discriminable	8. Be consistent	10. Movement compatibility	12. Hick-Hyman Law
	5. Exploit redundancy gain			13. Decision complexity advantage
	6. Avoid absolute judgement limits			14. Fitts's Law
				15. Provide feedback

Display Principles

Attention	Perceptual	Memory	Mental Model
1. Salience compatibility	5. Make displays legible (audible)	10. Knowledge in the world	14. Principle of pictorial realism
2. Minimize information access cost	6. Avoid absolute judgment limits	11. Support visual momentum	15. Principle of the moving part
3. Proximity compatibility	7. Support top-down processing	12. Provide predictive aiding	
4. Avoid resource competition	8. Exploit redundancy gain	13. Be consistent	
	9. Make discriminable		

Information Theory: Controls

- Taking action (response execution) after information processing
- Primarily involves selection and execution of responses
- Need to know how to support selection of responses and ultimately execution of control actions at system interfaces.

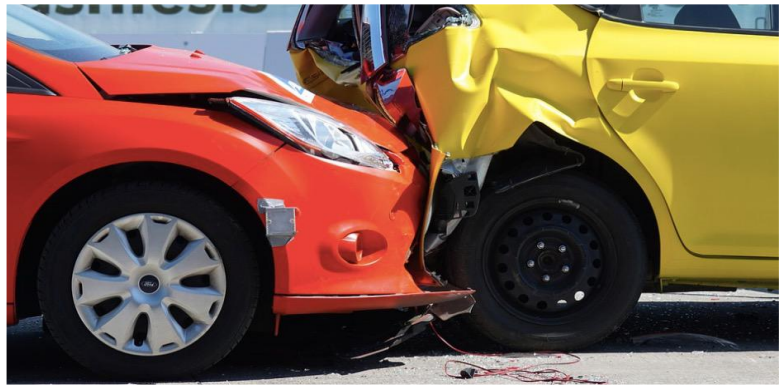


Human Factors Applications

- What are Orders of Control?
- Computer cursor control example:
  - Mouse cursor position is controlled with rate (first-order) controller
  - Mouse movement dictates rate/velocity of cursor motion:
    - If velocity of control device (mouse) is high, high gain is applied to cursor displacement (typically well above 1)
    - When control device moves slowly, low gain is applied (in some cases less than 1 for high accuracy cursor positioning)

Information theory and response selection and execution

- Human select more rapidly and accurately those actions that they expect to carry out than those are surprising to them



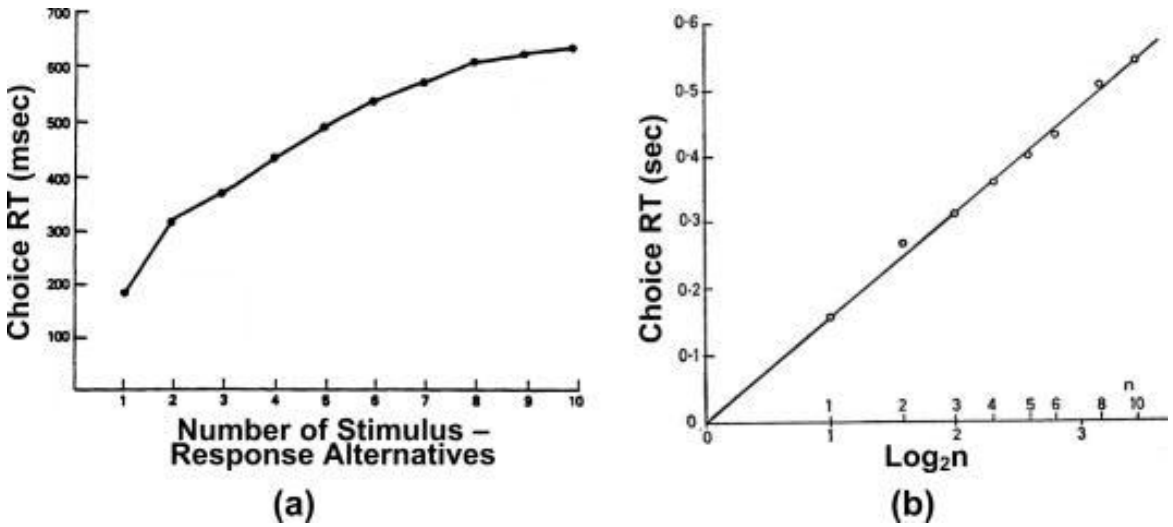
Do not expect the car in front of us to come to stop
We slow in perceiving visual field and in applying the brake

- Information theory mathematically explain the negative impact of uncertainty and complexity that human face in selecting and executing on response time.

Information theory and response selection and execution

- **<u>Hick-Hyman Law</u>** and Fitt's Law can quantify these impacts.
- Response selection time is dependent on number of control selections:
  - Fewer selections = faster response

Hick-Hyman Law



$$RT = a + b \log_2 N$$

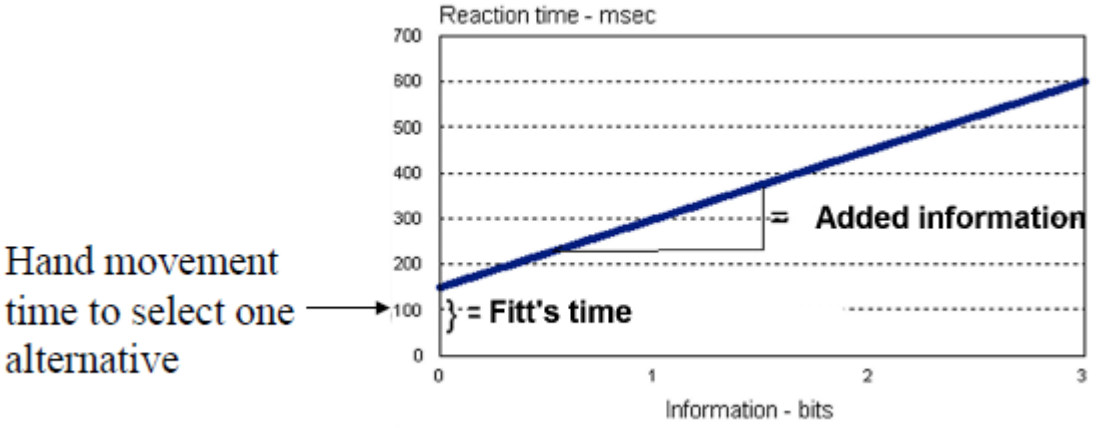
"The complexity of the controller should match the complexity of the control task"

RS12. Hick-Hyman Law

- Predicts time (RT) to select a response given the number of response alternatives (N)
  - $RT = a + b \log_2(N)$
- b = slope; additional processing time for each additional bit of information (determined empirically)
- a = sum of processing latencies unrelated to the amount of information
- Assumes that those choices are equally probable

Choice reaction time findings

- Reaction time (RT) increased as number of *equally likely* alternatives increased
- Hick converted number of choices (N) to bits of information (H): $H = \log_2 N$
- Found linear relation of RT against H:

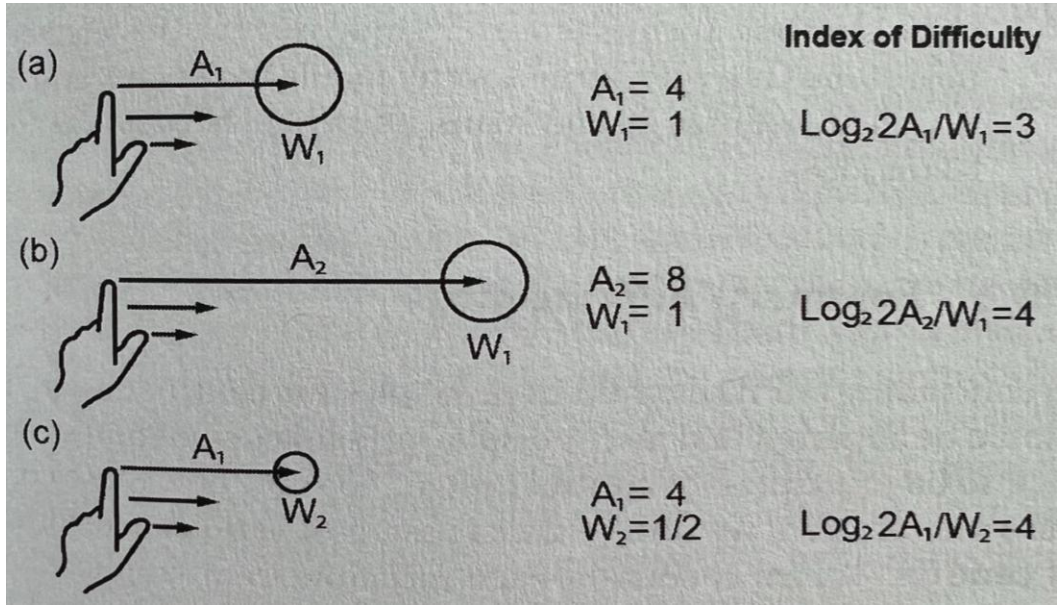


Fitts' time is minimum time to move hand to button (under no choice condition).

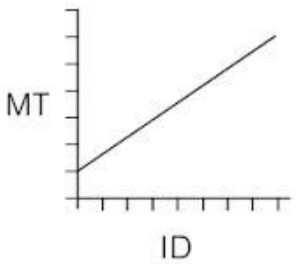
- Defined RT as function of H: $RT = a + b \log_2 N$

Information theory and response selection and execution

- Hick-Hyman Law and **Fitt's Law** can quantify these impacts.



Fitts' Law



$$MT = a + b^{*} ID$$

MT = Movement Time (*typically in milliseconds*)
a = intercept (*MT* where *ID* = 0)
b = slope
ID = Index of Difficulty

There is a linear relationship between MT (Movement Time) and the ID (Index of Difficulty)

When comparing different pointing devices a and b can be experimentally determined.

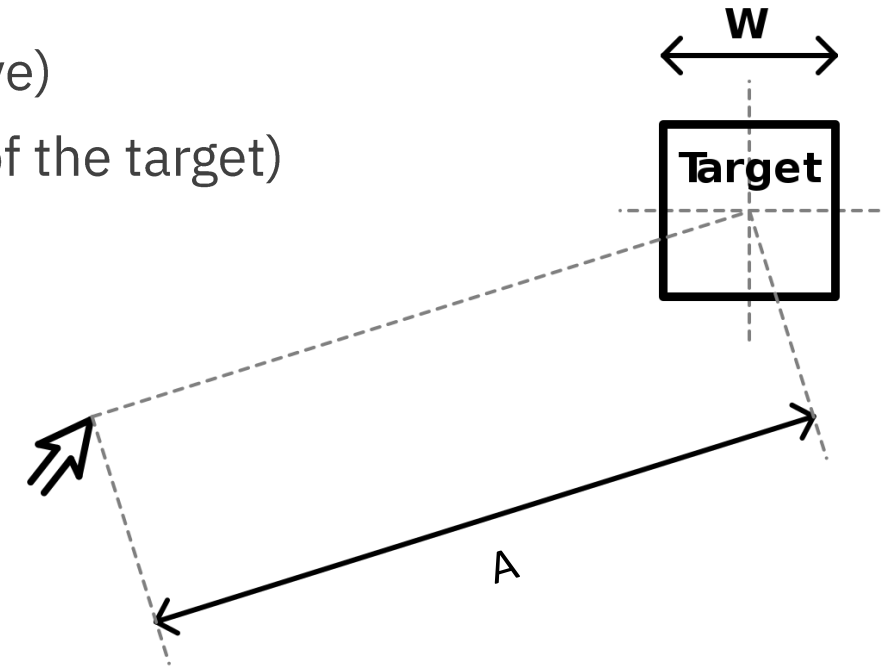
When predicting how quickly a target can be hit, a and b can be assigned a value.

"The complexity of the controller should match the complexity of the control task"

Information theory and response selection and execution

- Models the movement time required to reach a control
- Parameters of interest:
  - MT: Movement time
  - A: amplitude of movement (how far you need to move)
  - W: amplitude of target (how big the target is, width of the target)
  - a, b: constants

$$MT = a + b\,\underbrace{log_2(\frac{2A}{W})}_{\text{Index of difficulty}}$$



"The complexity of the controller should match the complexity of the control task"

Fitt's law

$$MT = a + b\,\underbrace{log_2(\frac{2A}{W})}_{\text{Index of difficulty}}$$

- Interpretation:
  - Intercept indicates individual differences in reaction time (unrelated to ID)
  - Larger and closer targets are acquired faster than smaller and farther targets

- Performing a target selection task is similar to "transmitting information through a channel"
- Amplitude of movement is the "signal", width is the "noise"
- Quantified by Throughput (bits / second)

$$\mathrm{TP} = \frac{\mathrm{ID}}{\mathrm{MT}}$$

<u>Uses of Fitts's Law:</u>

- Throughput can be compared between different pointing devices (ISO 9241-9)
- Fitts's Law can also be used to predict movement times

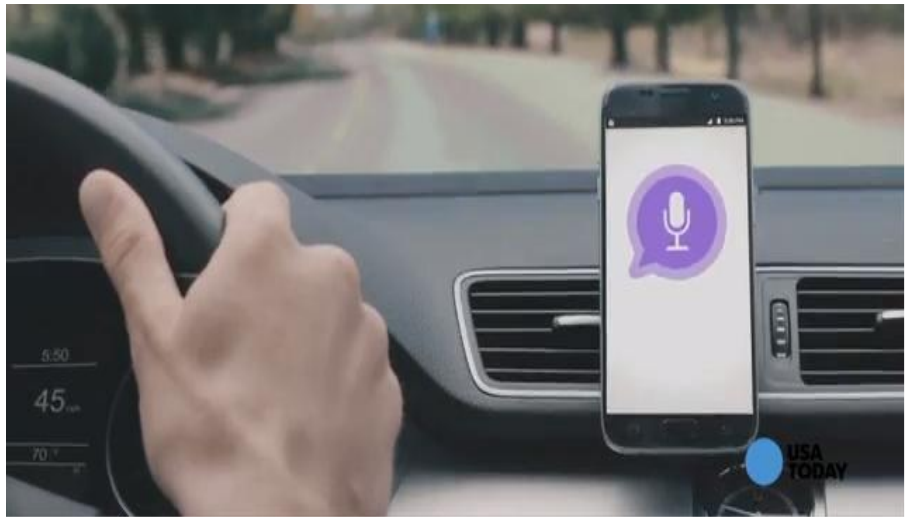
Control Principles

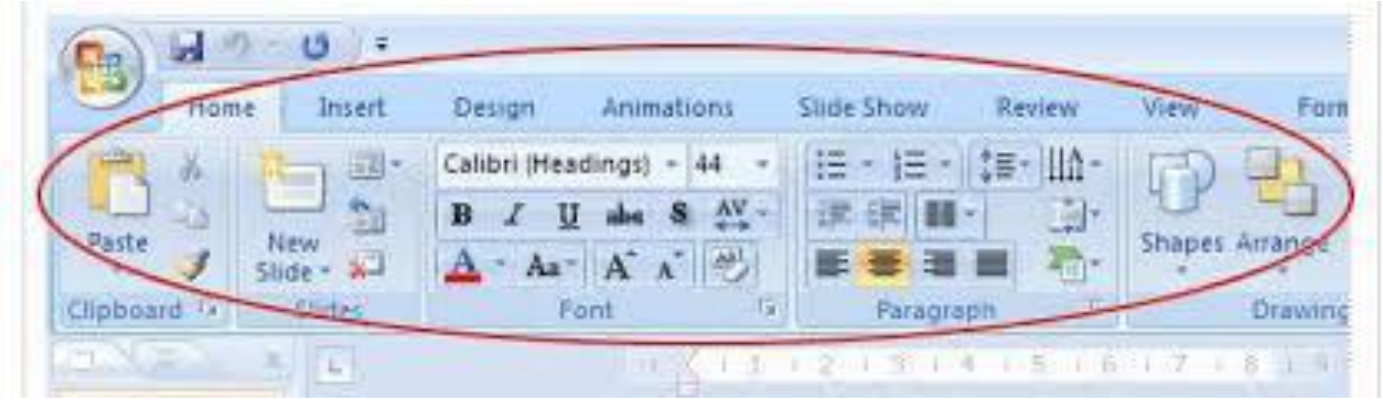
Attention	Perceptual	Memory	Mental Model	Response Selection
1. Proximity compatibility	3. Make accessible	7. Knowledge in the world	9. Location compatibility	11. Avoid accidental activation
2. Avoid resource competition	4. Make discriminable	8. Be consistent	10. Movement compatibility	12. Hick-Hyman Law
	5. Exploit redundancy gain			13. Decision complexity advantage
	6. Avoid absolute judgement limits			14. Fitts's Law
				15. Provide feedback

Control Principles

A. Attention

1. **Proximity compatibility**: The same as proximity compatibility for displays.
2. **Avoid resource competition**: When human are performing more than one task at the same time, multiple resource theory predicts a benefit of dividing the tasks across different mental resources.





Control Principles

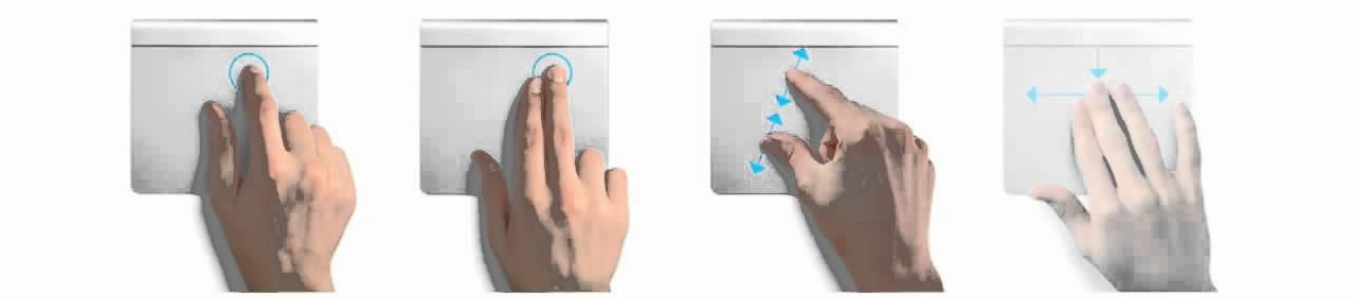
B. Perceptual

1. **Make accessible**: The controller should be easily reached.
2. **Make discriminable**: Identifying a particular control from an array of controls requires that they are discriminable.
3. **Exploit redundancy gain**: Redundancy gain can make controllers easier to identify and discriminate.
4. **Avoid absolute judgment limits**: Absolute judgement limits govern how many types of controls can be differentiated by anyone type of code such as color to no more than five to seven (Note: combining codes can expand this number).

Control Principles

C. Memory

1. **Knowledge in the world**: The actuation of the control should be reflected in the control itself, such that the position of toggle switch indicates its state or when the illumination of the button indicates its state or when the illumination of indicate a system is on.
2. **Be consistent**: Consistency makes it possible for people to apply skills from one situation to another, reducing errors and response time. Standardization should be considered for functions within a system as well as across systems (e.g. bad design could be location of light control between cars)



Control Principles

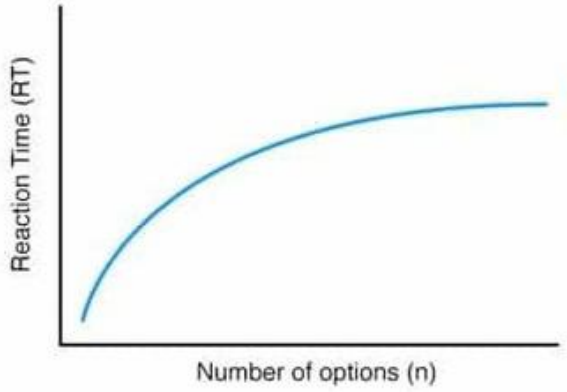
D. Mental Model

1. **Location compatibility**: Control location should be close to the entity being controlled or the display of the entity.
2. **Movement compatibility**: The direction of movement of a control should be congruent with direction both of movement of the displayed indicator and the system movement itself.

Control Principles

E. Response Selection

1. Avoid accidental activation
2. Hick-Hyman law
3. Fitt's law
4. Decision complexity advantage
5. Provide feedback

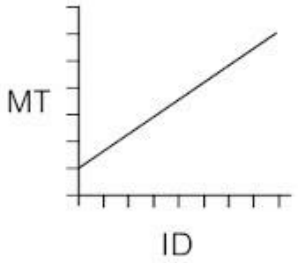


Hick's Law

RT = a + b log2 (n)

RT is the Reaction Time
(n) is the number of stimuli
"a" and **"b"** are constants

Fitts' Law



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RS11. Avoid accidental activation

Control forcing functions:

- Constraint on control operation
- Intended to prevent certain behaviors (usually errors) that would otherwise damage system
  - e.g., cannot shift automobile transmission from Park without depressing break
  - e.g., cannot shift automobile transmission into reverse when vehicle is in motion
  - e.g., putting weight on stairs causes the wheel lock to become engaged

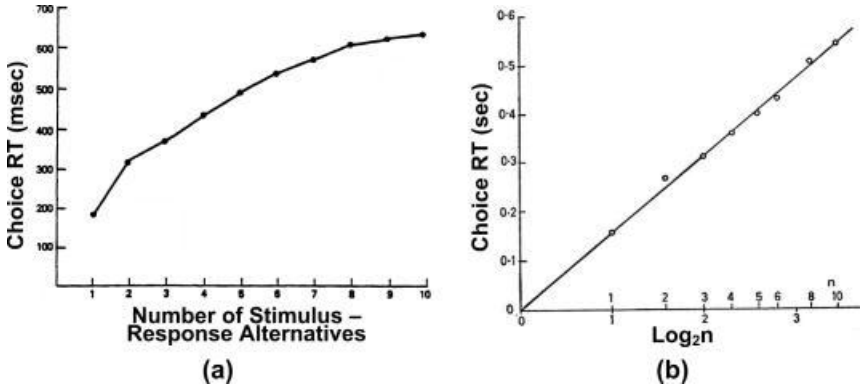
Human factors applications

- Constraint on control operation
- Intended to prevent certain behaviors (usually errors) that would otherwise damage system
  - e.g., cannot shift automobile transmission from Park without depressing break
  - e.g., cannot shift automobile transmission into reverse when vehicle is in motion
  - e.g. hand forklift cannot moved unless hand-brake is grasped and pull-lever is lowered

RS13. Decision Complexity Advantage

- What is the best way of going through a large number of options?

- More efficient to require a smaller number of complex decisions than many simple decisions
- Logarithmic relationship between number of stimulus and choice RT from Hick-Hyman Law



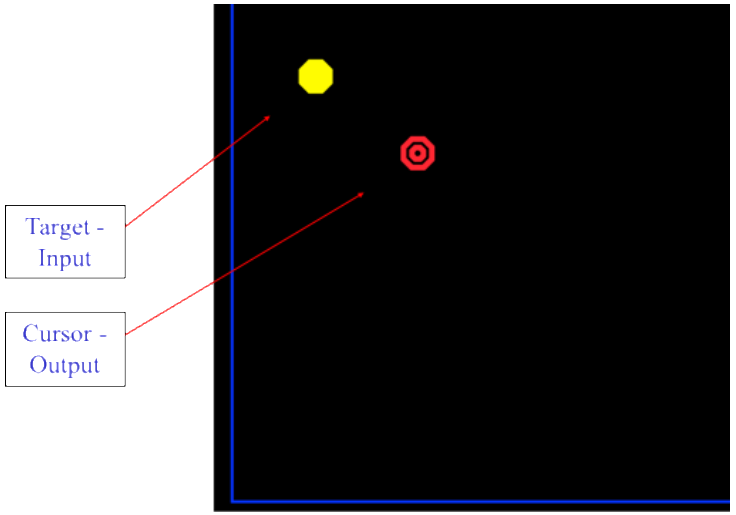
RS15. Provide Feedback

- System response to control / user input
- Multiple sensory channels can receive feedback:
  - Vision
  - Audition
  - Haptic sense
- Factors in real-time feedback effectiveness (e.g., warnings):
  - Clarity of presentation
  - Efficiency of presentation
  - Degree of user annoyance
  - User satisfaction
- Factors influencing use of feedback:
  - Human perceptual processing time (>200 ms)
  - Human information processing bandwidth
  - Human anticipation of information content or control options

Control Applications

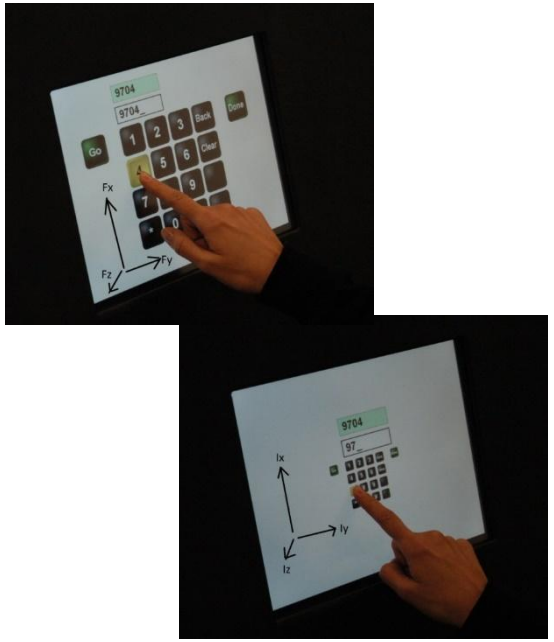
- Discrete

- Continuous



Principles for discrete control design

- Size:
  - Optimal control size provides optimal performance
- Labels:
  - Controls should be clearly labeled (at all times)
  - e.g., Backlit keyboard makes labels clear in dark environments
- Physical feel:
  - Must provide sensory feedback
  - e.g., Computer touch-point cursor control – Can feel resistance at stick
  - e.g., Computer touch-pad control – Textured to provide feel of finger motion across surface
  - e.g., Mouse button click as confirmation of control action



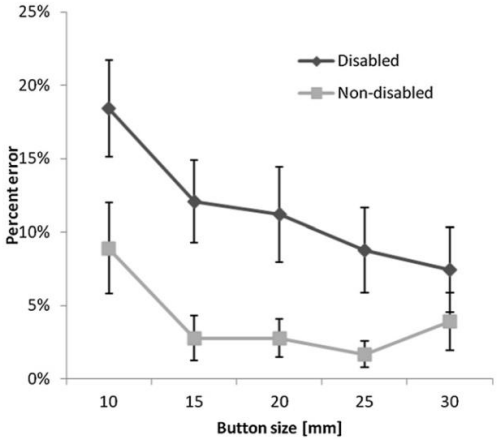


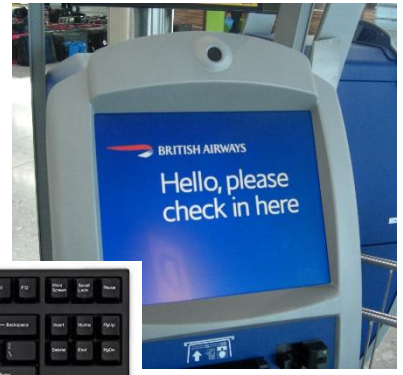
Fig. 3. Average error by button size and disability collapsed across gap (±1 Standard error).

Input devices

- Control actions are not limited to touch and motion
- Voice and visual control systems exist

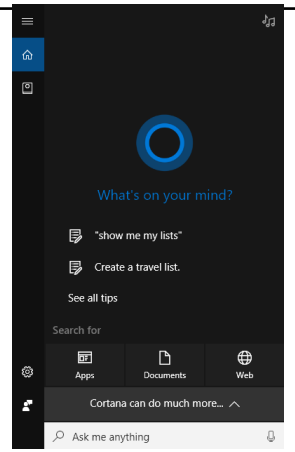
Touch

Touchscreens as alternatives to physical keyboards



Voice

Siri – Voice command recognition for phone use during multitasking



Visual

Figure 2. Pupil Pro r021. Front view of Pupil Pro (left image), shown with cable clip attached to user's collar. View from behind right (right image) with rotating clip attachment and flexible cable from headset to cable clip. Clip and lightweight and flexible cable enable greater freedom of head movement.

Pupil - Eye-tracking system for disabled user keying tasks

Motion / Gesture

Kinect – Motion tracking system for gesture recognition and commanding

UF

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LEADING THE CHARGE, CHARGING AHEAD